Chapter 1 - Introduction to Python

What is Python?

Python is a high-level, interpreted, general-purpose programming language created by Guido van Rossum and first released in 1991. It is famous for its clean and readable syntax, which makes it one of the best first languages for beginners. Python is used in web development, data science, artificial intelligence, automation, scripting and many other fields.

Key Features of Python

- Easy to read and write: code looks close to plain English.
- Interpreted: code runs line by line, so there is no separate compile step.
- Dynamically typed: you do not declare the type of a variable.
- Large standard library and a huge ecosystem of third-party packages (NumPy, pandas, FastAPI).
- Cross-platform: the same program runs on Windows, macOS and Linux.

Installing and Running Python

Download Python from python.org and check the installation from the terminal. You can type code directly in the interactive shell (REPL) or save it in a file ending in .py and run it.

```
python --version        # check the installed version
python                    # start the interactive shell
python hello.py           # run a script file
```

Your First Program

The print() function displays output on the screen. A line starting with # is a comment: Python ignores it, and programmers use comments to explain their code.

```
# My first Python program
print("Hello, World!")
name = input("What is your name? ")
print("Welcome to Python,", name)
```

Indentation Matters

Unlike many languages that use curly braces, Python uses indentation (usually 4 spaces) to define blocks of code. Wrong indentation causes an IndentationError.

How Python Runs Your Code

When you run a program, the Python interpreter first translates your source code into bytecode and then the Python Virtual Machine (PVM) executes that bytecode, one instruction at a time.